NO.:
DATE:

Uego, Francis Lise M. BSIT 2-1N OOP Activity #1

A. Variable identification type

Variable name	Data Type	Variable Type	Stored In
name	String	Instance Variable	Heap
grade	Int	Instance Variable	Heap
n	String	Local reference	Stack
g	Int	Local primitive	Stack
args	String	Local primitive	Stack
s1	Student	Local reference	Stack
s2	Student	Local reference	Stack
s3	Student	Local reference	Stack

B. Explanation

Statement	What happens	Memory Impact
Student s1 = new Student("John", 85);	- New student object is created	- s1 (reference variable) is created in the stack, and a new student object is allocated in the heap.
Student s2 = new Student("Maria", 90);	- Another student object is created	- s2 is created in the stack and a new Student object is allocated in the heap.
Student s3 = s2;	- Reference of s2 is copied to s3	- s3 copies memory address from s2. s3 is created in the stack, and no new heap allocation was done.
s1.grade = 95;	- Value of s1.grade is updated	- No changes in the stack. The grade field in the s1 student object has been updated in the heap.
s3.name = "Ana";	- Name reference in the shared object is updated.	- No changes in the stack. The name field in the s3 Student object now points to the address of "Ana" in the heap.

VICTORY

C. Output

John 95

Ana 90

Ana 90